

BONDING – COVALENT BONDING

KEY VOCABULARY

Covalent bond: a shared pair of electrons between two non-metal atoms.

Molecule: two or more atoms joined by covalent bonds.

Intermolecular force: weak force between molecules (not the bond).

Simple molecule: a small molecule like H₂O, CO₂ or Cl₂.

COMMON MISCONCEPTIONS

~~Covalent bonds are weak.~~

✓ Covalent bonds are strong; the forces between molecules are weak.

~~You break covalent bonds when boiling.~~

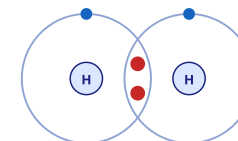
✓ You overcome weak intermolecular forces, not bonds.

~~Simple molecules conduct electricity.~~

✓ They have no free electrons or ions, so they don't conduct.

SHARED ELECTRONS (H₂)

A SHARED PAIR OF ELECTRONS



Each H shares one electron → both have a full shell (2)

1 DEFINE COVALENT BOND

State what is meant by a **covalent bond**. [1 mark]

2 WHICH ATOMS

State the type of elements that form covalent bonds. [1 mark]

3 LOW BOILING POINT

Explain why simple molecular substances have low boiling points. [3 marks]

Refer to intermolecular forces, not the covalent bonds.

4 CONDUCTIVITY

Explain why simple molecules do not conduct electricity. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Covalent bonds are ____ but the forces between molecules are ____.

shared

weak

transferred

strong

6 COUNT THE BONDS

Methane is CH₄. State how many **covalent bonds** it contains. [1 mark]